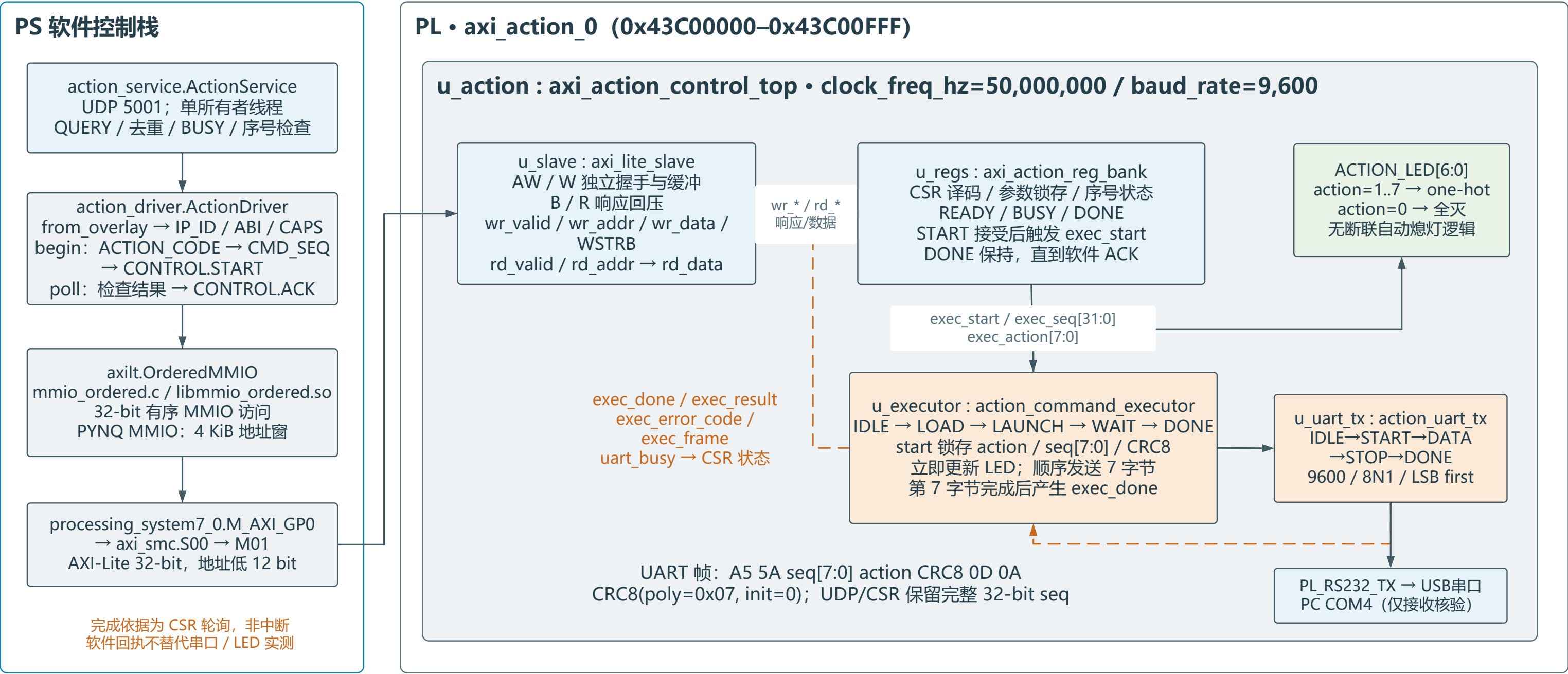


PS + PL 模块图：ACTL 寄存器、执行 FSM 与物理输出

实际层级：axi_action_0 / video_axi_action_uart_top / u_action:axi_action_control_top; 本动作位流没有 BLE 桥、BRAM 命令队列或模型核



ACTL v1 寄存器 (偏移相对 0x43C00000; 所有寄存器 32 bit)			
0x00 IP_ID = 0x4143544C	0x04 ABI_VERSION = 0x00010000	0x08 CAPS = 7	0x0C SCRATCH (读写测试)
0x10 ACTION_CODE (0..7)	0x14 CMD_SEQ (递增非零)	0x18 CONTROL: 1=START / 2=ACK	0x1C STATUS 位0/1/2/4: READY/BUSY/DONE/TX_BUSY
0x20 ACCEPT_SEQ	0x24 DONE_SEQ	0x28 LAST_ACTION	0x2C ERROR_CODE
0x30 EXEC_COUNT	0x34 LAST_FRAME: 00 CRC SEQ ACTION	0x38 OUTPUT_STATUS: bit0=1 / bit4=TX_BUSY	0x3C FRAME_COUNT (EXEC_COUNT 映射)